



ASSIGNMENT

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SECURITY FEATURES IN REAL-WORLD APPLICATIONS

- 1) Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security features/service required, such as confidentiality, integrity, authentication, authorization, and availability. The security mechanism is used to provide identified security features. The security protocol or technology used to implement the mechanism. Briefly explain how the mechanism/protocol provides the identified security feature.

Answer:

Application	Security Feature	Security Mechanism	Protocol/ Technology	How it provides Security
WhatsApp	Confidentiality	End-to-End Encryption (E2EE)	Signal Protocol	Encrypts messages that only the communicating users can read.
GitHub	Integrity & Authentication	Signed Commits and MFA	SSH, HTTPS/TLS, GPG, MFA	Helps authenticate the developers and allows users to verify the origin of signed commits.
Google Drive	Authorization	Access control	OAuth 2.0, IAM, ACLs	It determines which users can view, edit, download, or share files.
DigiLocker	Authentication & Authorization	Identity based authentication	Aadhar/OTP-based mechanisms, TLS	Helps authenticate users and securely provide access to digital documents.

Telegram	Confidentiality	Encryption	MTPROTO / Secret Chats	Secret chats provide end-to-end encryption between participants.
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2) For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering security attacks. Mention the mechanisms and protocols involved.

Answer:

Application	Confidentiality Controls	Integrity Controls	Availability Controls
WhatsApp	End-to-End Encryption using Signal Protocol protects the messages from being read and encrypted transport using TLS.	Authenticated encryption and key verification help prevent/detect tampering.	Redundant/distributed infrastructure keeps messaging services available.
GitHub	HTTPS/TLS protects source code and credentials during transmission.	SHA-1/SHA-256 hashes & GPG/SSH signed commits help verify that code/commits haven't been altered and establishes the signer's identity.	Backups, replication and DDoS protection help keep repositories accessible.
Google Drive	TLS, encryption at rest and access permissions protect stored and transmitted files.	Version History helps identify and recover unwanted changes to files.	Replication and redundant cloud infrastructure keep files available even if systems fail.

DigiLocker	HTTPS/TLS and access control protect stored and transmitted files.	Digital signatures and PKI help verify the authenticity and integrity of digital documents.	Cloud redundancy backups and recovery systems help maintain access to documents.
Telegram	MTPROTO encryption; E2EE in Secret chats protect message content.	Cryptographic authentication mechanisms in MTPROTO help protect against message modification.	Distributed cloud infrastructure and redundancy help maintain service availability.

Mechanisms & Protocols Involved

- ***WhatsApp***
 - End-to-End Encryption (E2EE) – Signal Protocol, AES (Advanced Encryption Standard), Curve25519 Algorithm.
- ***GitHub***
 - Cryptographic Hashing & Digital Signatures – SHA (Secure Hash Algorithm)- 256, GPG (GNU Privacy Guard), SSH (Secure Shell).
- ***Google Drive***
 - Authorization & Access Control – OAuth 2.0, IAM (Identity and Access Management), ACLs (Access Control List)
- ***DigiLocker***
 - Digital Signatures & Authentication – PKI (Public Key Infrastructure), X.509 Certificates.
- ***Telegram***
 - Encryption & E2EE – MTPROTO, AES-256, Diffie-Hellman.